Project Executable Files & Deployment

Template: Project Executable Files and Deployment Instructions	**Maximum Marks: 2 Marks**
Date: 15 March 2026	**Team ID:** FV-AI-2026-X1
Project Name: FoodVerse AI	**Track:** AI/ML & Generative AI

Production Cloud Deployment URLs (No Local Setup Required):

Deployment Layer	Hosting Provider	Live Access URL	Status
Live Frontend Web App	Vercel Global Edge CDN	https://foodverse-ai-six.vercel.app/	LIVE / PRODUCTION
Live Backend API Gateway	Render Cloud Web Services	https://foodverse-backend.onrender.com/	ACTIVE
API Documentation (Swagger)	FastAPI Interactive Docs	https://foodverse-backend.onrender.com/docs	INTERACTIVE
Source Code Repository	GitHub (Public)	https://github.com/AbhiragMishra/foodverse-ai	UP-TO-DATE

Local Execution & Startup Scripts:

Developers and evaluators can launch the full-stack system locally with single-click automation.

Executable Script	Target Platform	Command to Run	Action Performed
start_foodverse.bat	Windows 10 / 11	.\start_foodverse.bat	Automatically initializes backend virtualenv, starts FastAPI on port 8000, and launches React Vite frontend on port 5173.
start_foodverse.sh	Linux / macOS	chmod +x start_foodverse.sh && ./start_foodverse.sh	Bash automation script starting asynchronous backend and frontend services with graceful exit handling.
test_edge_cases.py	Cross-Platform (Python)	python backend/test_edge_cases.py	Executes automated 9-point unit test suite validating edge cases, corrupted media, and extreme diets.

Core Executable File Manifest:

File Name	Directory Path	Role in System	Key Dependencies
main.py	backend/main.py	FastAPI root server & middleware configuration	fastapi, uvicorn, python-dotenv
food_recognition.py	backend/routers/food_recognition.py	Plate identification & nutritional estimation router	groq, PIL, pydantic
freshness_detector.py	backend/routers/freshness_detector.py	Computer vision produce decay analyzer	opencv-python-headless, numpy
hygiene_monitor.py	backend/routers/hygiene_monitor.py	Commercial kitchen video frame auditor	opencv-python-headless, groq
kitchen_assistant.py	backend/routers/kitchen_assistant.py	Recipe generation and culinary voice Q&A;	groq, pydantic
meal_planner.py	backend/routers/meal_planner.py	Clinical metabolic meal planning engine	pydantic, Mifflin-St Jeor logic
App.jsx	frontend/src/App.jsx	Main React application workbench & UI panels	react, vite, canvas API